Simplifying Virtual Research Environments with SimpleVM: A Comprehensive Platform to boost Research and Training Productivity

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SimpleVM

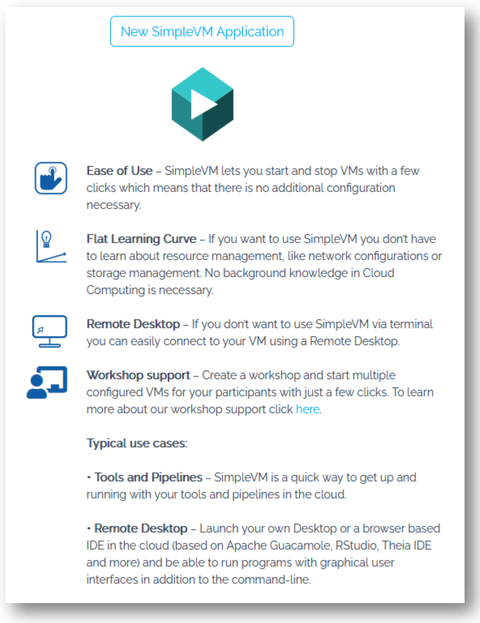
Simple access to computational resources for challenging bioinformatics analyses is often difficult for researchers to obtain. We therefore previously introduced SimpleVM in the federated de.NBI Cloud, based on OpenStack. It is now available as a stand-alone application that can be deployed on any OpenStack cloud environment and uses authentication and authorization through Life Science AAI or alternative AAI services, coupled to EduGAIN. SimpleVMs support basic data-processing use-cases, but can also serve sophisticated, cluster-based computing, GPU-accelerated machine learning and workshop scenarios.

Project Application

LS LOGIN

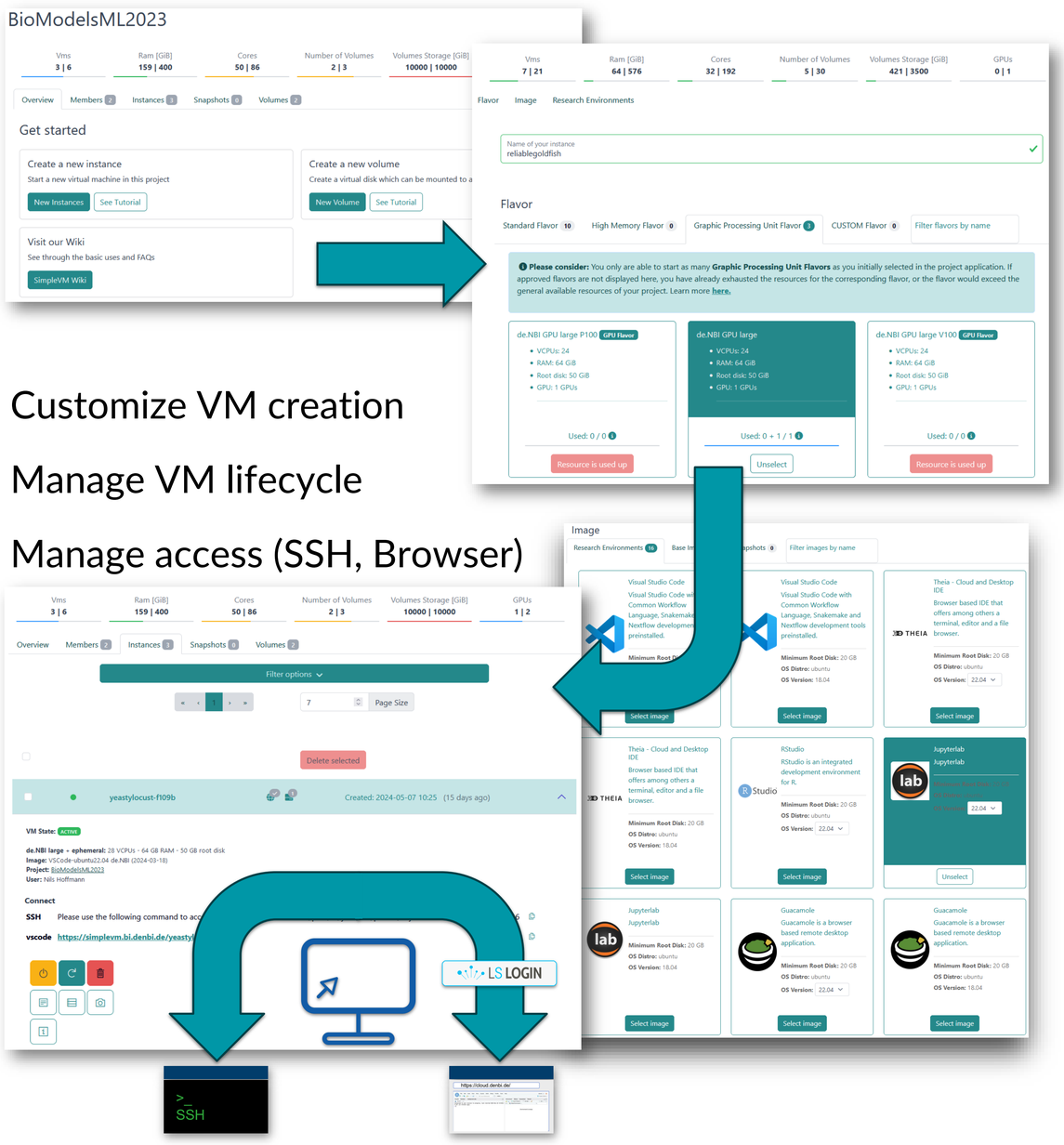
eduGAIN

- Applicants (PI or delegate) log in via Life Science Login
- Applicants select a SimpleVM project type, enter a project name, a description of project goals, estimated required resources, tools planned to be used and references to methods & desired lifetime
- Application review panel decides to proceed and allocate project or reiterates with applicants



VM Creation & Management

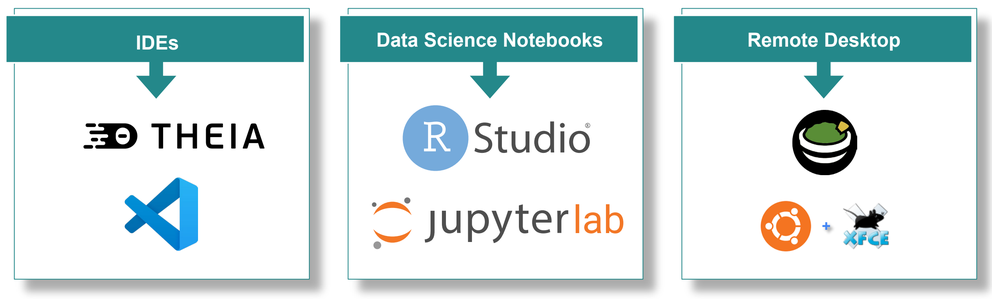
- Select from Standard, High Mem, GPU and Special Flavors, depending on your requirements and workload
- Create and manage block storage volumes for your VMs



- Customize VM creation
- Manage VM lifecycle
- Manage access (SSH, Browser)

Research Environments

- Pre-configured, based on Packer recipes (GitHub)
- Customize with base image + tools installed with Bioconda



Training Support

Workshop

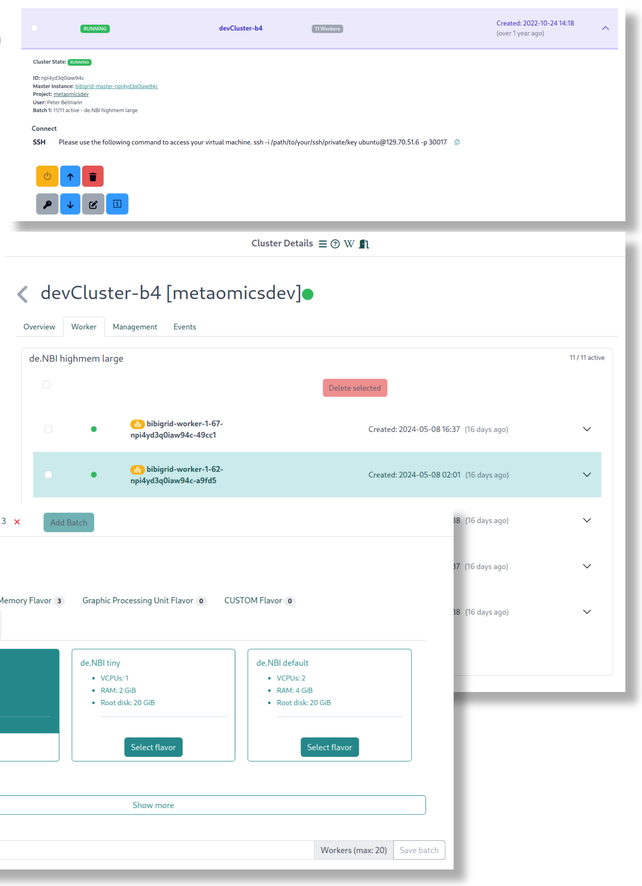
- Easy setup of training participant VMs from image or snapshot
- Management of VMs and assigned participants by project administrators
- Successfully used in de.NBI Cloud User Meetings 2023 (Unix Shell, Research Environments), de.NBI Summer School 2023 (Unix Shell), de.NBI Spring School 2024 (Metagenomics) and other trainings

CLOUD USER MEETING 23 BIELEFELD

SLURM Grid Cluster

Cluster

- Based on BiBiGrid (SLURM) cluster
- Automatic bootstrapping
- Flavors and images of both Master server and worker can differ
- Batches support different worker configurations
- Current work on autoscaling
- Compatible with Nextflow



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